INCH-POUND
A-A-59776A
November 29, 2011
SUPERSEDING
A-A-59776
June 22, 2005

COMMERCIAL ITEM DESCRIPTION

GEAR, FIRE PROTECTIVE

The General Services Administration has authorized the use of this commercial item description, for all federal agencies.

1. SCOPE. This commercial item description provides the requirements for a structural fireman's coverall used by U.S. Navy firefighters.

2. CLASSIFICATION. The Fire Protective Gear (FPG) shall conform to the following:

2.1 Sizes. The FPG shall be available in the eighteen (18) sizes listed below.

Small-Short	Small-Regular	Small-Long
Medium-Short	Medium-Regular	Medium-Long
Large-Short	Large-Regular	Large-Long
X-Large-Short	X-Large-Regular	X-Large-Long
XX-Large-Short	XX-Large-Regular	XX-Large-Long
XXX-Large-Short	XXX-Large-Regular	XXX-Large-Long

3. SALIENT CHARACTERISTICS.

3.1 Design and construction. The FPG (see figure 1) shall be a one-piece garment with a snap-in removable liner system. The FPG shall be a raglan sleeve design with side and crotch gussets. The FPG and all layers of the FPG shall be individually graded per size. The finished FPG and all its sub-components shall be in accordance with the latest version of the National Fire Protection Association (NFPA) 1971 standard. The FPG shall also meet the additional requirements as specified herein for use in a shipboard environment. The FPG shall have the following features:

a. Front closure – The front closure shall have a zipper and hook and pile flap closure in accordance with NFPA 1971.

b. Elastic waistband – There shall be a 1¼ inch wide elastic waistband sewn into the outer garment starting and stopping at the front closure. The waistband shall be covered by a two-inch wide piece of outer garment material.

c. Leg zipper – There shall be a leg zipper at the bottom of each leg along the outside seam. The zipper shall not be less than 18 inches long when measured from top of top stop to the bottom of the bottom stop and allow the leg opening to be fully open while in the unzipped position.

d. Sleeve cuffs – The sleeve cuff of the outer shell shall be a partially gathered elastic cuff with a hook and pile tab closure.

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e. Cargo pockets with flap closures - There shall be a cargo pocket with flap closure located on side of each hip below the waist and above the knee area, (see figure 1). The cargo pocket shall have grommet(s) at the bottom of the pocket to allow for drainage. The pocket may be manufactured out of one or more pieces of the outer shell material. The cargo pocket shall be three dimensional- two sides, one front and one bottom panel. The internal dimensions of pocket shall be 8 ± 1/8 inches wide by 8 ± 1/8 inches long by 2 ± 1/8 inches deep at any point within the pocket. The flap closure shall be at least the same width as the pocket and secured to the pocket by using a hook and pile closure tab extending the width of the pocket. The hook section shall be placed on the flap closure. The flap closure has to be fully secured when the pocket is in its extended state."

f. Radio cargo pocket with flap closure – There shall be a radio cargo pocket with flap closure located on the left side of the chest, (see figure 10). The radio cargo pocket shall have grommet(s) at the bottom of the pocket to allow for drainage. The pocket may be manufactured out of one or more pieces of the outer shell material. The radio cargo pocket shall be three dimensional- two sides, one front and one bottom panel. The internal dimensions of pocket shall be-3¾ ± 1/8 inches wide by 7½ ± 1/8 inches long by 2½ ± 1/8 inches deep at any point within pocket. The flap closure shall be at least the same width as the pocket and-secured to the pocket by using hook and pile closure tab extending the width of the pocket. The hook section shall be placed on the flap closure. The flap closure has to be fully secured when the pocket is in its extended state."

g. Retro-reflective fluorescent trim – The retro-reflective fluorescent trim shall be two inches in width and meet the placement requirements of Option 1 in NFPA 1971

h. Reinforcement material – The elbows, complete bottom half of the cargo pockets, and the complete bottom half of the radio cargo pocket shall use outer shell material as reinforcement material. The reinforcement material used for the knee and leg cuffs shall be in accordance with 3.2.6. The reinforcement material for the knee area shall not be less than a 9 by 9 inch area and shall be located to cover the wearer's knee while kneeling. The reinforcement material for the leg cuffs shall have a minimum height of ¾ inches.

i. Knee/shoulder heat protection – The knee and shoulder area shall be provided with additional material to meet the requirements in 3.2.2. The shoulder area material shall measure at least 4 inches in length along the crown of the shoulder and extending down from the crown on both the front and back of the garment at least 4 inches. The crown of the shoulder shall be the uppermost line of the shoulder when the garment is laying flat on an inspection surface with all closures fastened. The knee area protection material shall correspond to the dimensions and location of the knee reinforcement area as specified in 3.1.h.

j. Over-the-thumb wristlets – There shall be an over-the-thumb wristlet attached to the liner on each sleeve. Each wristlet shall have a minimum un-stretched length of 8 inches and shall contain a thumbhole that has a minimum diameter of 1 inch. The wristlet length shall be measured from the farthest end of the attachment point to the liner to the end of the wristlet. The wristlets shall meet the requirements of NFPA 1971.

k. Stand up collar – There shall be a stand up collar that encircles the neck and is compliant with NFPA 1971. The collar shall have a minimum height of three inches. The collar shall have a circumference that allows the collar to be closed around the neck without discomfort to the user. The collar shall be secured by a hook and pile tab closure.

3.1.1 Color. The color of the outer shell shall be natural. The knee/leg cuff reinforcements shall be gray or gold in shade and be the same color. The color of the retro-reflective fluorescent trim shall be yellow.

3.1.2 Materials and components. All fabrics and batting used in the construction of the outer shell, liner, and wristlets shall be manufactured from inherently flame-resistant fibers. All other materials used in the construction of the garment, such as zipper tapes, collar linings, elastic, hook and pile fasteners, padding, reinforcements, interfacing, bindings, hanger loops, emblems, labels, patches, and trim, shall meet the requirements of NFPA 1971.

3.1.3 Finished dimensions. The finished dimensions of the garments shall be in compliance with Tables I and II.

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TABLE I. FPG sizes and corresponding finished dimensions (inches).

Size	Back rise (A)	Front rise (B)	Chest (C)	Hip (D)	Sleeve length (E)			Waist (F)	Inseam (G)		
					Short	Regular	Long		Short	Regular	Long
Small	38	32½	46	48	31½	33½	35½	32	27	29	31
Medium	39	33½	50	52	33	35	37	36	27	29	31
Large	40	34½	54	56	33½	35½	37½	40	27	29	31
X-Large	41	35½	58	60	34	36	38	44	27	29	31
XX-Large	42	36½	62	64	34½	36½	38½	48	27	29	31
XXX-Large	43	37½	66	68	35	37	39	52	27	29	31
Tolerances	+1/-½	+1/-½	+1/-½	+1/-½	+1/-½	+1/-½	+1/-½	+1/-½	+1/-½	+1/-½	+1/-½

NOTES:

1. All measurements shall be taken with the lining removed, and the outer shell placed flat on a clean surface, with front opening fully closed (except for front rise measurement). See figures 2 through 10 for measurement locations.
2. The center point of the crotch gusset shall be determined by folding the garment (knee pads outward) in half and super imposing one leg over the other leg. Lay legs out on flat surface ensuring crotch gusset is flat and smooth. Mark outer most point of crotch gusset (see figure 2.a). Open garment and measure crotch gusset width. Divide width in half and mark garment (see figure 2.b). Extend both lines to be perpendicular to each other. The intersection of the two lines will be identified as the center point of crotch gusset (see figure 2.c).
3. The front centerline of the garment shall be the center of the garment at the inner seam of the collar to the center point of the crotch gusset. The back centerline shall be from the midpoint of the collar line between the back seams to the center point of the crotch gusset.
4. The back rise shall be measured from the center point of the crotch gusset along the back centerline of the garment to the seam where the collar is attached to the garment (see figure 3).
5. The front rise shall be measured from the center point of the crotch along the front centerline of the garment to the inner seam where the collar is attached to the front of the garment (see figure 4).
6. The chest shall be measured across the front from folded edge to folded edge, crossing the intersections of seams on the left and right side body gussets (see figure 5). The measurement shall be doubled.
7. The hip shall be measured across the garment (from folded edge to folded edge) 8 inches below the centerline of the waistband; the measurement shall be doubled (see figure 6).
8. The sleeve length shall be measured from the garment's back centerline and collar seam intersection to the bottom of the un-stretched sleeve cuff (see figure 7).
9. The waist measurement shall be taken with the waistband fully extended from the left side of the waistband to the right side of the waistband; the measurement shall be doubled (see figure 8).
10. The inseam is the measurement from the center point of crotch gusset to the bottom of the leg (see figure 9).

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TABLE II. Wrist and leg opening dimensions (inches).

Size	Wrist extended	Wrist relaxed	Leg opening
Small	14½	12	21
Medium	14½	12	21½
Large	14½	12	22
X-Large	14½	12	22½
XX-Large	14½	12	23
XXX-Large	14½	12	23½
Tolerance	±¼	±¼	±½

NOTES:
1. The relaxed measurement on the wrist shall be measured with sleeve cuff laying flat. Beginning at the folded corner, measure over to the end of the folded flat cuff; the measurement shall be doubled.
2. The extended measurement on the wrist shall be measured with sleeve cuff laying flat and completely extended. Beginning at the folded corner, measure over to the end of the folded flat cuff; the measurement shall be doubled.
3. The leg opening shall be measured with the leg unzipped, laying flat and open; measure from one edge of the leg cuff to the opposite side edge.

3.2 Performance requirements.

3.2.1 Thermal. The FPG shall have a minimum Thermal Protective Performance (TPP) rating of 40 when tested in accordance with NFPA 1971.

3.2.2 Conductive/compressive heat resistance (CCHR). The knee and shoulder shall have a minimum heat transfer index of 100 for the knee and 65 for the shoulder areas of the FPG. The testing shall be in conformance with the following test procedure:

a. Samples shall measure 6 by 6 inches, ±¼ inch.

b. Samples for the knee and shoulder areas shall be representative of the composites used in those areas of the FPG.

c. A total of six specimens shall be tested. Three specimens from the knee area and three specimens from the shoulder area shall be tested.

d. Specimens shall be preconditioned in accordance with standard textile laboratory conditions specified in ASTM D1776.

e. Sample specimens shall be tested in accordance with ASTM F1060 with the modifications specified herein.

f. Specimens shall be tested using an exposure temperature of 536 °F (280 °C).

g. The pressure applied during the test shall be 2±0.2 pounds per square inch (psi) [0.008±0.0008 kilogram per square centimeter (kg/cm^2)] for shoulder and 8±0.8 psi (0.032±0.0032 kg/cm^2) for knee areas.

h. The CCHR rating shall be the time in seconds to achieve a temperature rise of 43±3.6 °F (24±2 °C) using a starting temperature of 68±9 °F (20±5 °C) for all specimens. The time shall be measured to the nearest tenth of a second.

i. The individual CCHR rating for each specimen shall be reported. The average CCHR rating shall be calculated and reported.

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j. If an individual result from any test set varies more than ±8 percent from the average results, the results from the test set shall be discarded and another set of specimens shall be tested.

3.2.3 Outer shell.

3.2.3.1 Weight. The outer shell fabric shall have a maximum weight of 7.9 ounces per square yard (oz/sq yd) when tested in accordance with ASTM D3776, Option C.

3.2.3.2 Strength. The outer shell fabric shall have a 22-pound retained strength after heat exposure. The testing shall conform to the following test procedure:

a. A sample of the outer shell by itself shall be placed on the thermal performance tester in accordance with NFPA 1971.

b. The test shall be performed with the following exceptions:

(1) Only standard laboratory preconditioning shall be performed in accordance with ASTM D1776.

(2) A 17.5-second exposure shall be performed.

(3) The recording of results shall not be accomplished.

(4) The testing shall be conducted on five specimens.

(5) The only weight on the sample shall be the sensor block; no additional weight shall be used.

c. Following the 17.5-second exposure, the exposed samples shall be tested for burst strength in accordance with ASTM D6797.

d. The average shall not be lower than the requirement as stated above.

3.2.3.3 Weight Loss. The outer shell fabric shall have a maximum weight loss of 4.0 percent after abrasion testing. The testing shall be in accordance with ASTM D3884 and shall be performed with the following exceptions:

a. All samples shall be weighed to the one hundredth place prior to abrasion.

b. Samples shall be run for 500 cycles with H18 surface grit wheels. Each wheel, with auxiliary weights, shall weigh 500 grams. A total weight of 1000 grams shall be applied to the specimen.

c. Taber abrasion machine shall be run with vacuum in the off position. Loose particles shall be air-blown cleaned from each sample after testing, and then re-weighed and percent weight loss calculated.

d. Cleaning of each wheel shall take place after each individual specimen is run 500 cycles. Periodic cleaning of wheels during the 500 cycles may be conducted as needed to maintain a relatively clean wheel surface. Cleaning shall be conducted with a small stiff brush, applying minimal to no pressure to the running wheel.

e. Upon completion of each fabric sample set (N=5), each H18 surface grit wheel shall be resurfaced in accordance with ASTM D3884. Utilize resurface disc material of 150-grit. Resurfacing duration shall be 25 cycles. After resurfacing, clean the abrasion wheels with a vacuum to remove dust. The resurfacing disc should be replaced after 50 total cycles.

3.2.4 Liner. The FPG liner shall consist of a moisture barrier and a thermal barrier. The maximum weight of the liner shall be 8.3 ounces per square yard when tested in accordance with ASTM D3776, Option C. The moisture barrier shall have a water vapor diffusion resistance no greater than 650 seconds per meter (s/m) when tested in accordance with ASTM F2298, Method A. The specimens shall be tested in the cell for a minimum of 1 hour. The thermal barrier liner face cloth shall have a minimum of 25 percent by weight filament fibers when tested in accordance with AATCC 20A.

3.2.5 Retro-reflective fluorescent trim. The coefficient of retro reflection for the retro-reflective fluorescent trim after heat exposure shall be a minimum value of 100 candela per lux per square meter ($cd/lux/m^2$) when tested as follows:

a. The specimens shall be preconditioned by performing the heat and thermal shrinkage resistance test in accordance with NFPA 1971. Thermal shrinkage shall not be measured.

b. The specimens shall be tested for retro reflection after being preconditioned in accordance with NFPA 1971.

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3.2.6 Reinforcement material. The material shall contain a high solids polymer face coating for protection against abrasion. The maximum weight shall be 20.0 ounces per square yard when tested in accordance with ASTM D3776, Option C. The maximum thermal shrinkage in either direction shall be 1.0 percent when tested in accordance with NFPA 1971. The maximum vertical flammability shall be no more than 1 second after flame and a maximum char length of 1.0 inch when tested in accordance with ASTM D6413.

4. REGULATORY REQUIREMENTS. The offeror/contractor is encouraged to use recovered materials to the maximum extent practicable, in accordance with paragraph 23.403 of the Federal Acquisition Regulation (FAR).

5. PRODUCT CONFORMANCE PROVISIONS.

5.1 Product conformance. The products provided shall meet the salient characteristics of this CID, conform to the producer's own drawings, specifications, standards, and quality assurance practices, and be the same product offered for sale in the commercial marketplace. The manufacturer shall provide documentation that shows the Fire Protective Gear is compliant with the latest version of NFPA 1971 standard and certified third party test data showing that the Fire Protective Gear conforms to the requirements specified in this CID prior to contract award.

6. PACKAGING. Preservation, packing, and marking shall be as specified in the contract or order.

7. NOTES.

7.1 Source of documents.

7.1.1 ASTM. ASTM standards are available from ASTM International, 100 Barr Harbor Dr., P.O. Box C700, West Conshohocken, PA 19428-2959 or online at www.astm.org.

7.1.2 FAR. The Federal Acquisition Regulation may be obtained from the Superintendent of Documents, U.S. Government Printing Office, Washington, DC 20402 or online at www.acquisition.gov/far.

7.1.3 NFPA. National Fire Protection Association (NFPA) standards are available from NFPA, 1 Batterymarch Park, Quincy, MA 02169-7471 or online at www.nfpa.org.

7.1.4 AATCC. American Association of Textile Chemists and Colorists test methods are available from AATCC, P.O. Box 12215, One Davis Drive, Research Triangle Park, NC 27709-2215 or online at www.aatcc.org.

7.2 Ordering data. The contract or order should specify the following:

a. CID document number and revision.
b. Product conformance provisions.
c. Quantity required.
d. NSN or size ordered.

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7.3 National stock numbers (NSNs). The following is a list of NSNs assigned that correspond to this CID. The list may not be indicative of all possible NSNs associated with the CID.

NSN	Item	NSN	Item
4210-01-468-5505	Small/Short FPG	4210-01-468-5599	X-Large/Short FPG
4210-01-468-5528	Small/Regular FPG	4210-01-468-5671	X-Large/Regular FPG
4210-01-468-5539	Small/Long FPG	4210-01-468-5673	X-Large/Long FPG
4210-01-468-5545	Medium/Short FPG	4210-01-468-5677	XX-Large/Short FPG
4210-01-468-5551	Medium/Regular FPG	4210-01-468-5679	XX-Large/Regular FPG
4210-01-468-5552	Medium/Long FPG	4210-01-468-5681	XX-Large/Long FPG
4210-01-468-5562	Large/Short FPG	4210-01-468-5683	XXX-Large/Short FPG
4210-01-468-5565	Large/Regular FPG	4210-01-468-5689	XXX-Large/Regular FPG
4210-01-468-5568	Large/Long FPG	4210-01-468-5691	XXX-Large/Long FPG

7.4 Key words.

Firefighting coverall

Firefighting ensemble

Firefighting suit

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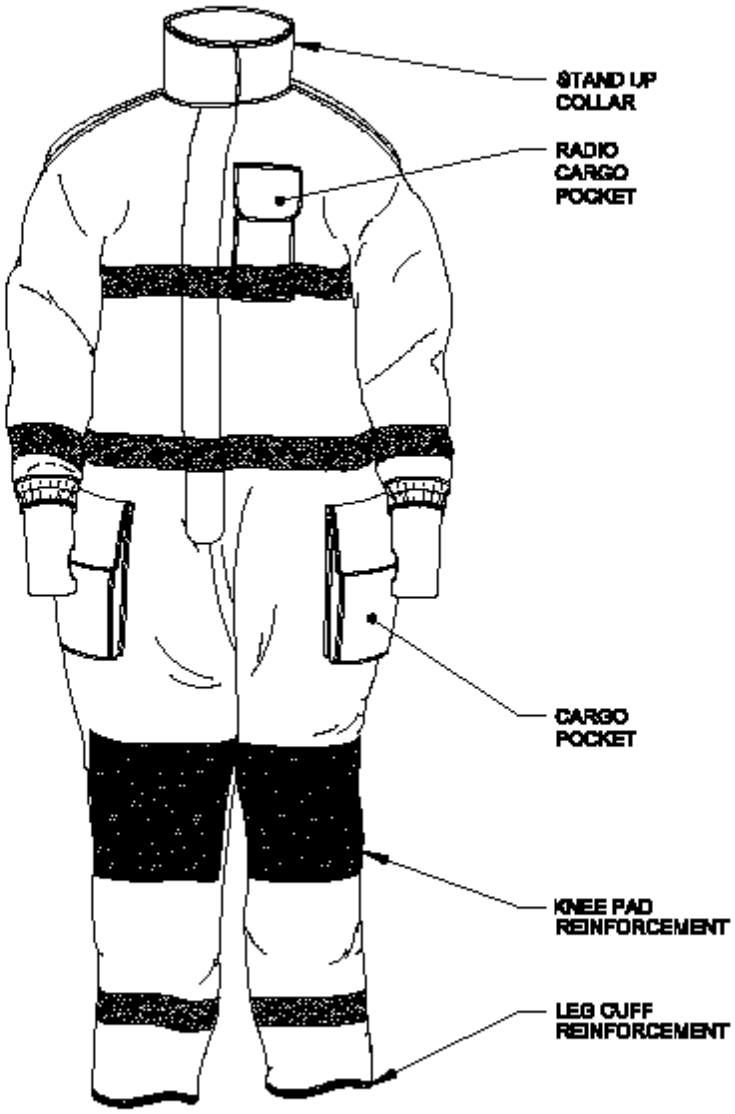


FIGURE 1. Front view of the Fire Protective Gear (FPG).

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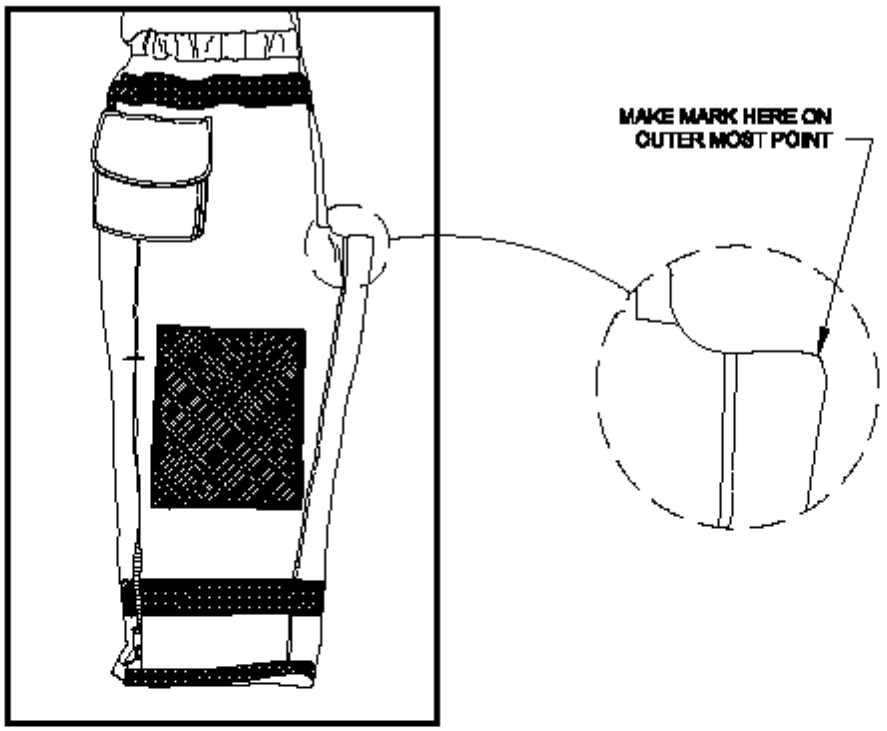


FIGURE 2A

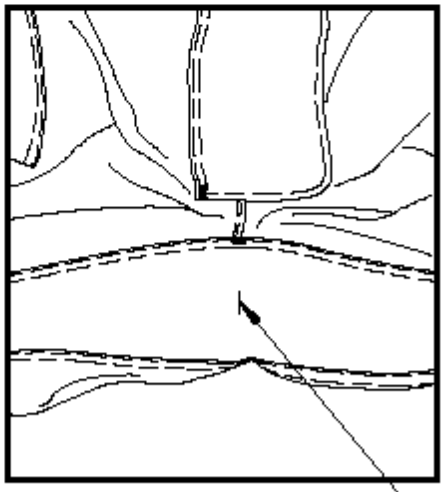


FIGURE 2B

PREVIOUSLY MARKED LINE FROM FIGURE 2A

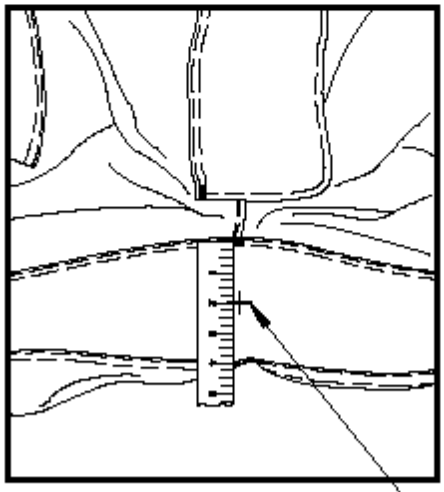


FIGURE 2C

MARK LINE IN CENTER OF CROTCH GUSSET
THAT INTERSECTS LINE SHOWN IN FIGURE 2B

FIGURE 2. Center point of crotch gusset.

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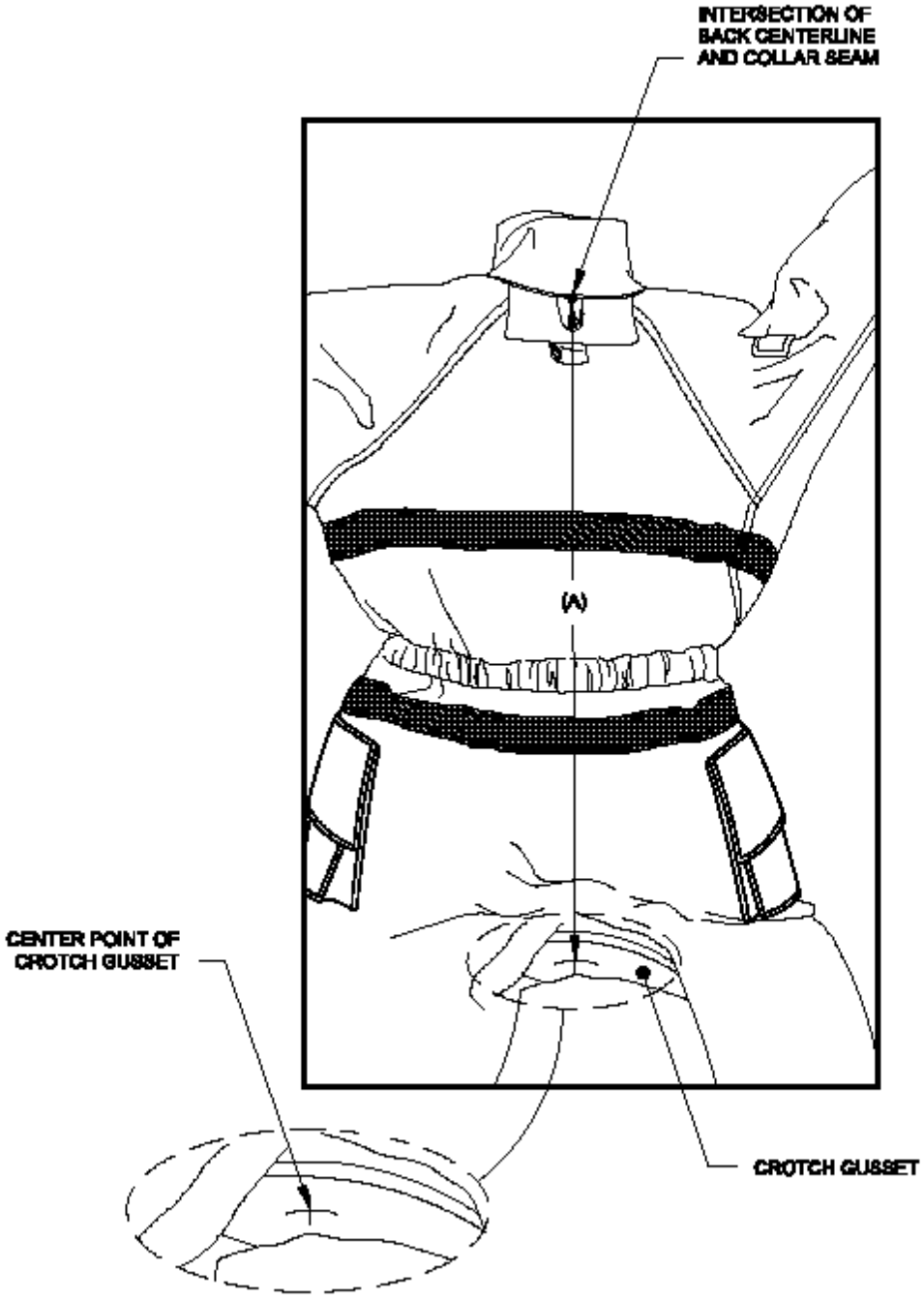


FIGURE 3. Back rise measurement (A).

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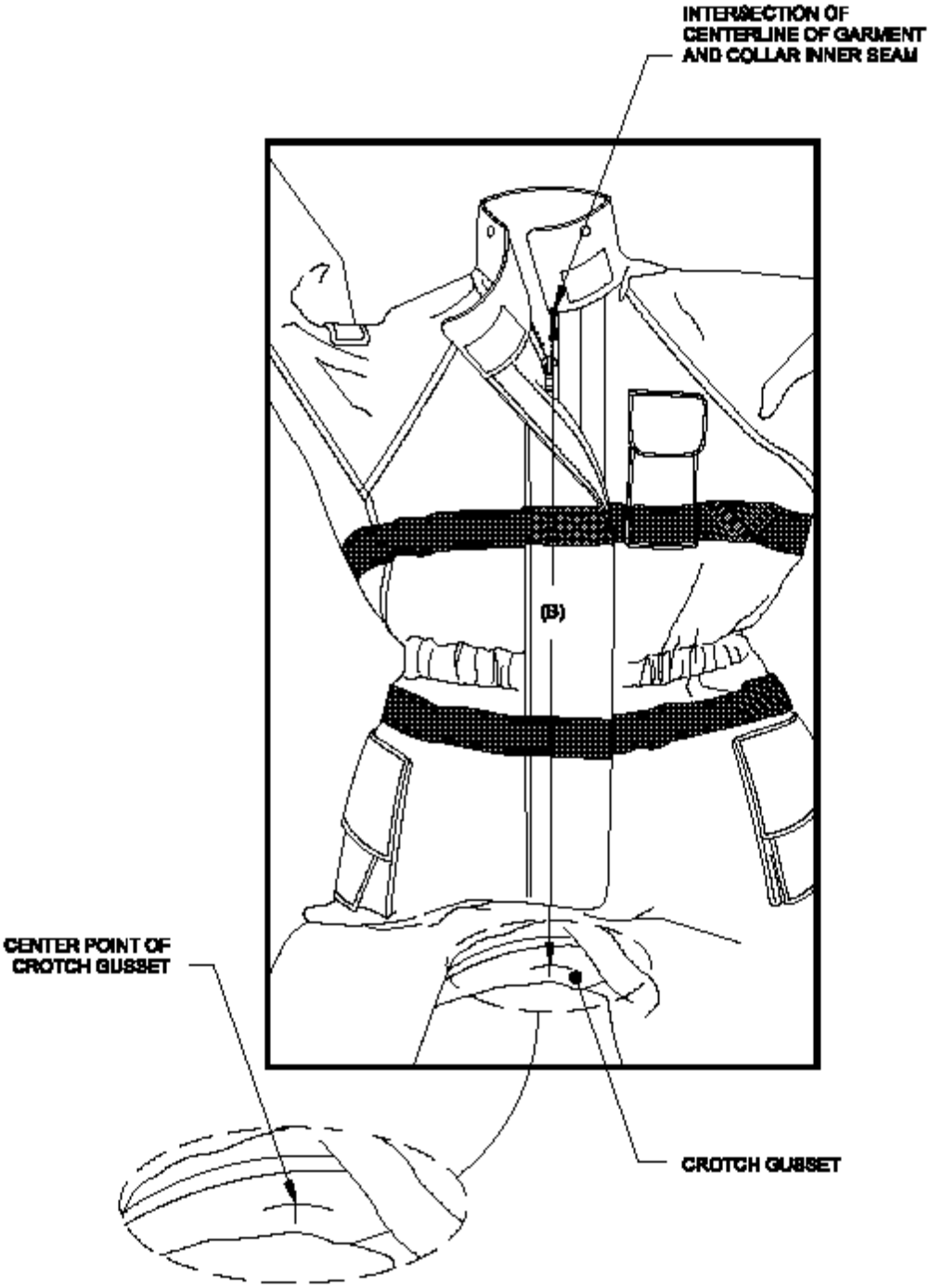


FIGURE 4. Front rise measurement (B).

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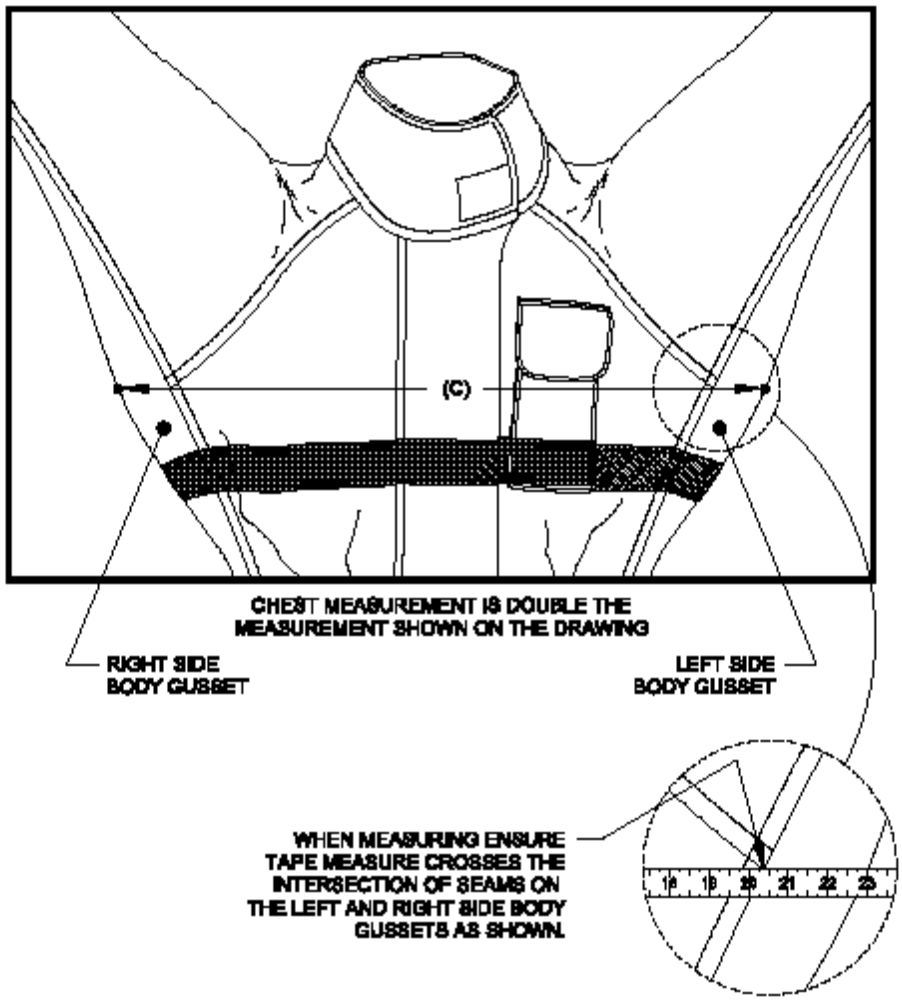


FIGURE 5. Chest measurement (C).

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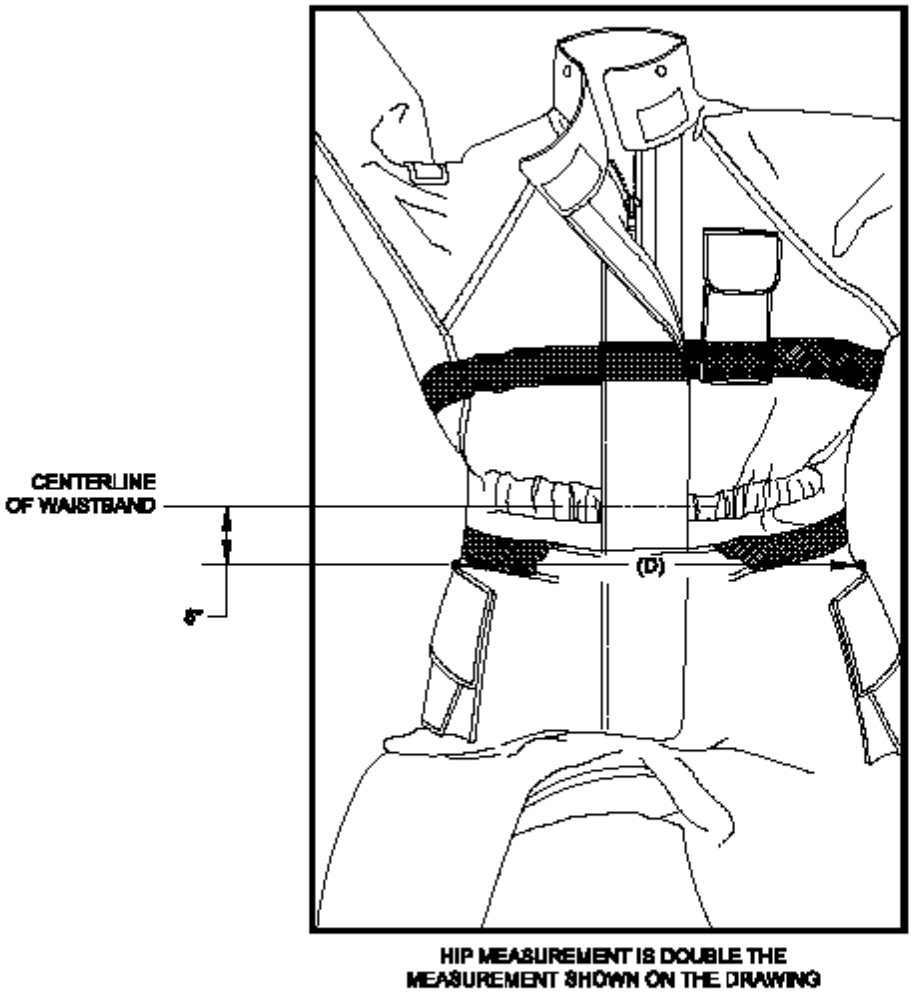


FIGURE 6. Hip measurement (D).

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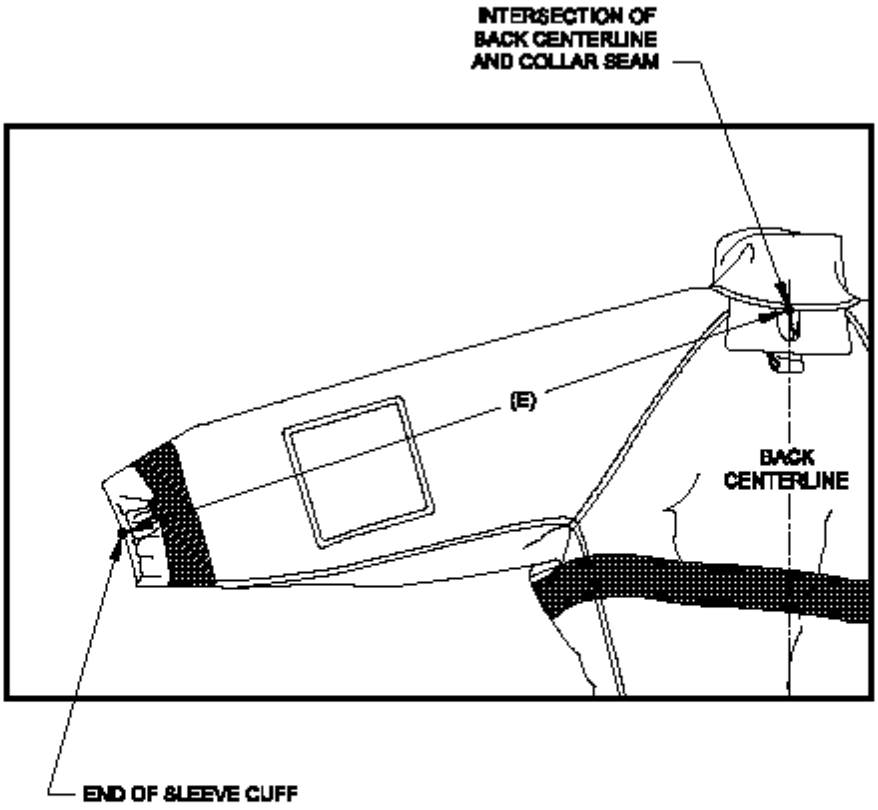


FIGURE 7. Sleeve measurement (E).

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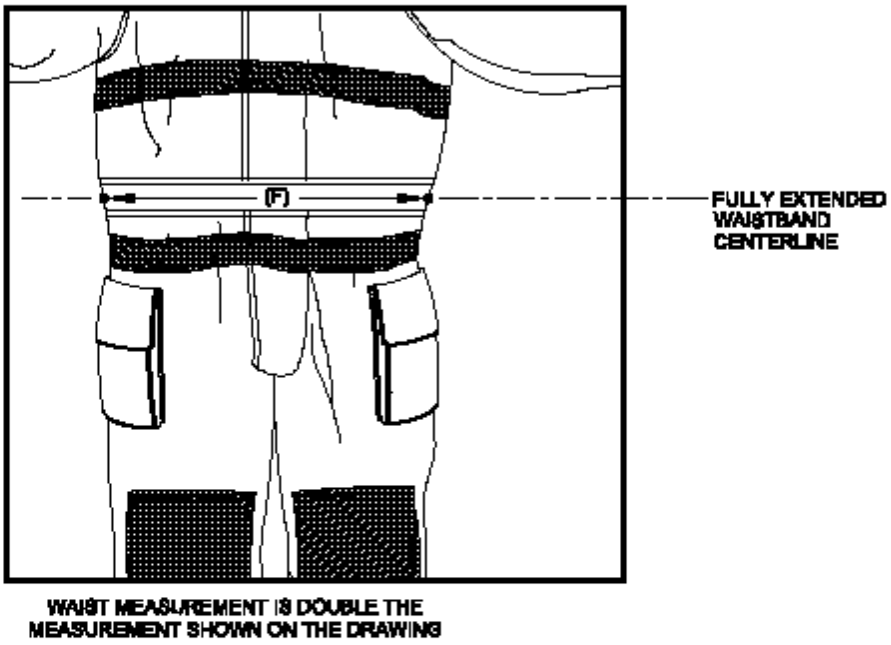


FIGURE 8. Waist measurement (F).

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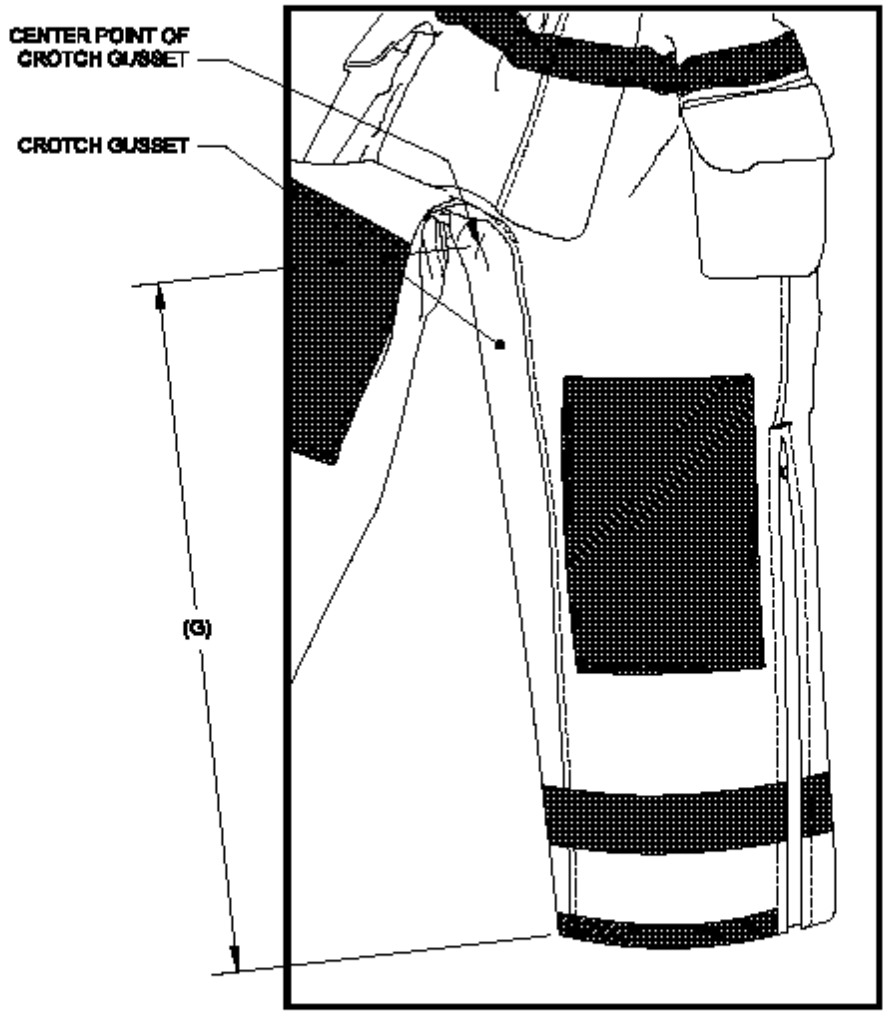


FIGURE 9. Waist measurement (G).

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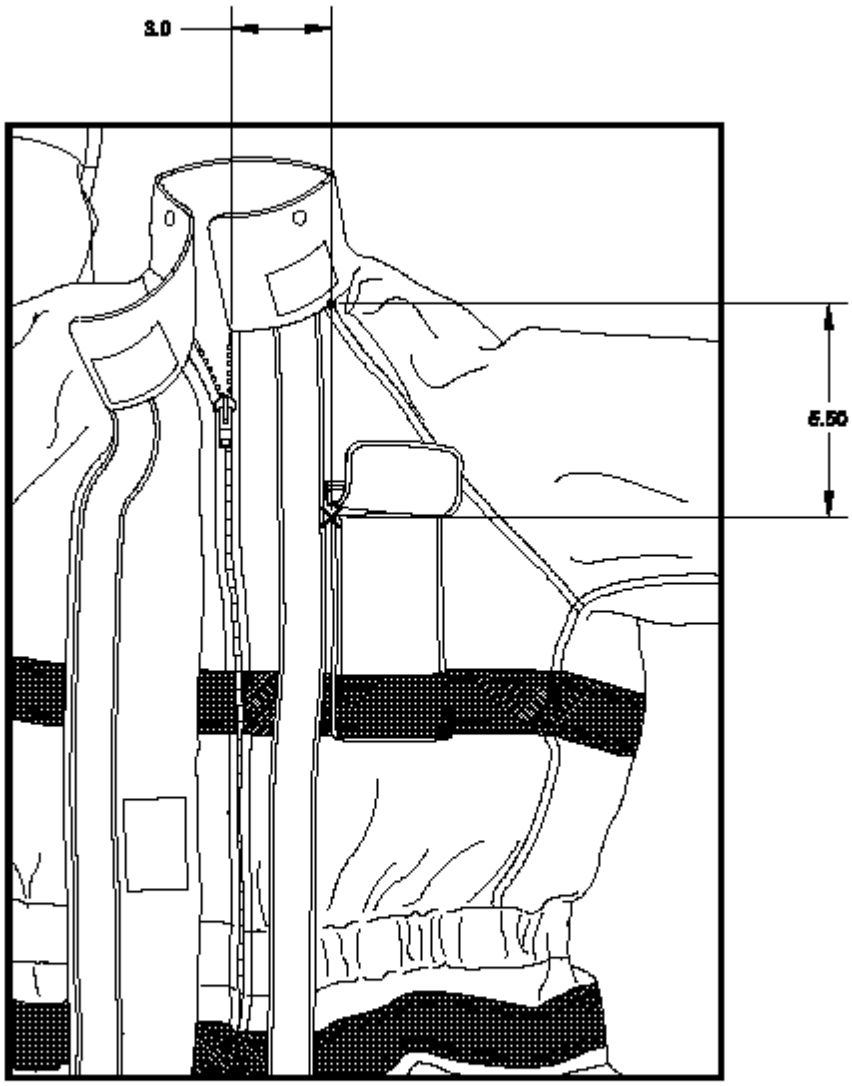


FIGURE 10. Radio cargo pocket location.

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MILITARY INTERESTS

Custodians:
Navy – SH
Air Force – 03

Review Activities:
Navy – AS, NU
Air Force – 84, 99
DLA – IS

Preparing Activity:
Navy – SH
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